Resolution Geometry: The Case for Engagement

Twelve Silent Failures of Established Tools, Seven Dilemmas, and the Ledger of Partial Visibility

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Abstract

Resolution Geometry (RG) has no place to stand except the cracks between established theories. It has to show those cracks before anyone has a reason to look. This paper is written for the reader who intends to decline. It does not ask for agreement with a new axiom. It asks the reader to check twelve exact computations and seven implications from theorems they already accept.

Silent failures. Each of the twelve is a case in which a standard tool gives a wrong answer without warning. Examples:

- two experiments with the same Fisher information at every parameter that are not the same experiment;
- a limit test along every ray that passes while the function is discontinuous;
- a ratio and its denominator treated as independent;
- a unitary total evolution whose reduced channel cannot be inverted;
- a metric component taken as a prediction;
- an equivalence-principle argument that misses half the deflection of light.

Every witness is exact or symbolic, and each is caught by a named RG object.

Dilemmas. Each of the seven has the form: if you accept a cited theorem, then you accept an RG consequence, unless you reject a specific checkable claim. The theorems are Čencov's, Lovelock's, Petz's, the nerve theorem, Hardt's, the hole argument with Choquet-Bruhat–Geroch, and data processing with Kalman observability.

Objections answered. We answer the objections RG must expect: arbitrariness, redundancy, coincidence, circularity, tautology, lack of novelty, softness, unfalsifiability, selection and parasitism. Each answer is a pointer to a proof, a frozen computation or a no-go. We list what would refute RG, and we separate what is classical from what is claimed.

Conclusion. RG is not an add-on to established theory. It is the bookkeeping that those theories' own theorems force, once their residue is no longer left implicit.

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1 One premise everyone already accepts

Definition 1.1 (Records, fibres, ignorance)**.** A *protocol* assigns to each preparation a law of records. Two preparations are *indistinguishable* for the protocol if their record laws coincide. The equivalence class of a preparation is its *fibre*. A quantity is *determined* by the protocol iff it is constant on fibres. Otherwise it carries *ignorance*.

The tautology objection, answered in advance. "A science uses only what its records distinguish" is a tautology, and it is meant to be one. A tautology everyone accepts is the strongest possible premise. RG's content is not this sentence. It is the set of theorems about fibres:

- they are nontrivial in every established theory examined;
- they come in four provably distinct kinds;
- the only admissible local geometry on them is forced (Čencov from descent);
- that geometry is provably insufficient beyond the local scale.

None of these is a tautology. Each is checked below.

2 Twelve silent failures

A *silent failure* is a case in which a standard tool, used in its standard way, returns a wrong answer and gives no warning. Each witness is exact or symbolic (battery `rg_silent_failures.py`).

	Standard tool	Exact witness of failure	RG object that catches it
S1	Fisher information as the information of the experiment	Bernoulli($\sin^2\frac{\theta}{2}$) and two i.i.d. Bernoulli($\sin^2\frac{\theta}{2\sqrt{2}}$) both have Fisher $\equiv 1$. Their Hellinger affinities at separation 1 are 0.877583 and 0.880122, so neither is a randomization of the other	global comparison (rooms, deficiency)
S2	the local metric selects the divergence	same Hessian ($f''(1) = 1$) but KL, $\frac{1}{2}\chi^2$ and $2\,\text{Hellinger}^2$ equal 0.534, 0.595 and 0.544 at fixed laws	task-declared finite loss

	Standard tool	Exact witness of failure	RG object that catches it
S3	limits along rays	every ray of $xy^2/(x^2+y^4)$ gives 0; the curve $x=y^2$ gives $\frac{1}{2}$	definable-curve crossing criterion
S4	the Fisher kernel is the unidentifiable direction	$p_\theta = (\frac{1}{2}+\theta^2, \frac{1}{2}-\theta^2)$ has $F(0)=0$ but $\mathrm{TV}=\theta^2>0$	hidden versus null; wall rate $\lambda=\frac{1}{4}$
S5	zero local loss everywhere means sufficiency	two components with laws $(.2,.8)$ and $(.8,.2)$, with the record erased: zero defect everywhere, but no common recovery	global sufficiency; components are rooms
S6	different labels mean independent rows	$\mathrm{Cov}(X, Y/X) = -Y\sigma_X^2/X^2 = -0.0043$ for neutron f_{13} and f_{14}/f_{13}	covariance follows records, not labels
S7	fit the samples you have	with $3M=3$ samples, frequency 1.1 reproduces a frequency-0.7 trace exactly	minimal probe $3M+1$, abstention
S8	a unitary total evolution means recoverable information	complete dephasing sends $\|+\rangle$ and $\|-\rangle$ to the same $I/2$	joint reversal versus reduced recovery
S9	metric components are physical	at the same event $g_{tt}=-1$ in proper-time gauge and $g_{\eta\eta}=-a^2=-1.717$ in conformal gauge; $a(\tau)$ is gauge-free	relational readout; spacetime as representative
S10	the equivalence principle fixes light bending	box (energy channel only): $0.876''$; full GR (with structure): $1.751''$	energy versus structure channels
S11	a map is fixed by its action on states	real QM: identity and transpose agree on every local state but differ on $(I+0.2\,\sigma_y\otimes\sigma_y)/4$	extension completeness
S12	L^2 distance between distributions	merging two outcomes raises it from $\frac{3}{8}$ to $\frac{1}{2}$	descent forces Fisher

What the list establishes. Each established framework that a reader may work in contains at least one silent failure that is fixed by typing the residue. S6 was committed by the author of this programme, in an earlier version of the GR–QM protocol paper, and caught by exactly this typing. The failures are not exotic. They arise in ordinary data analysis, estimation, relativity and quantum information.

3 The language is forced, and it is not redundant

RG uses four kinds of not-knowing. The redundancy objection says these are relabelings of known notions. The answer is that they are pairwise distinct, and each distinction is witnessed.

Proposition 3.1 (Four distinct kinds of ignorance)**.** *RG distinguishes four kinds of ignorance:*

- Null*: never visible to any record (gauge).*
- Hidden*: invisible instantaneously, but visible in histories or extensions.*
- Room*: a global equivalence class not captured by any local quantity.*
- Obstruction*: a global extension class, such as H^1 or a π_k class.*

These are pairwise distinct:

- *null versus hidden: Heisenberg observability, where r_y is recovered and r_x is null;*

- *hidden versus room: S1 and S5, where every local quantity agrees but the global classes differ;*
- *room versus obstruction: L131, where a fibre type is fixed but a winding class is not;*
- *local degeneracy versus any of them: S4, where Fisher-null is not null.*

Consequence. A framework that lacks these four types must conflate at least two of them, and the corresponding silent failure then occurs. The vocabulary of ignorance, obstruction and blindness is therefore not decoration. It is the minimal typing that prevents S1–S12.

4 Seven dilemmas

Dilemma 1 (Čencov)**.** *If you accept Čencov's theorem, Fisher is the only admissible local geometry. Since the lossless re-descriptions are exactly congruent splittings, this follows from descent alone. S1, S2, S4 and S5 then show that it does not determine global equivalence, finite loss, walls or global sufficiency. So the residue must be typed elsewhere. To decline, reject Čencov or refute an exact witness.*

Dilemma 2 (Spacetime)**.** *If you accept the hole argument and Choquet-Bruhat–Geroch uniqueness up to isometry, every GR prediction is a relational observable, and spacetime points are a gauge representative (S9). To decline, exhibit a GR prediction that changes under an isometry.*

Dilemma 3 (The field equation)**.** *If you accept Lovelock's theorem and the two-loop R^3 divergence (Goroff–Sagnotti), the Einstein equation is the unique law* inside a declared class*, and the declaration that selects it is violated by its own quantum completion. So the universality of GR is not provable. To decline, exhibit a UV completion that reduces to GR with vanishing R^3.*

Dilemma 4 (The quantum metric)**.** *If you accept Petz's classification, quantum descent keeps every monotone metric. Their relative spread reaches* 16.6 *on noncommuting tangents, so the metric must be declared by a task. To decline, exhibit a task-free quantum principle that selects one member.*

Dilemma 5 (Global from finite)**.** *If you accept the nerve theorem and the aliasing witnesses (S7, and the exact holonomy threshold* $\max(\pi/N, \pi/2 - 2\alpha)$*), then every global claim from finite samples requires a declared coverage, and below it abstention is mandatory. To decline, exhibit an estimator that beats the alias bound.*

Dilemma 6 (Tameness)**.** *If you accept Hardt's theorem and curve selection, every tame experiment has finitely many fibre types and a decidable crossing criterion, while the* $\sin(1/\theta)$ *family has infinitely many walls. So tameness must be declared. To decline, prove finiteness for a non-definable family.*

Dilemma 7 (Hidden versus null)**.** *If you accept data processing and Kalman (Krylov) observability, then history recovers hidden values and never recovers null ones, and the Fisher kernel at a point confuses the two (S4). To decline, show that a Fisher-null direction is always null.*

5 RG is not soft: it forbids

A soft framework can fit anything. RG forbids specific, common claims.

- It forbids inferring equivalence of experiments from equal Fisher fields (S1).
- It forbids treating supernova data as a test of $D = 4$: flat DGP fits Pantheon+ as well as ΛCDM, with $\Delta\chi^2 = -0.44$.

- It forbids treating the spectral scale of neutron levels as an equivalence-principle test, since the scale has rank one.
- It forbids reading the continuous holonomy from finite node data. Three admissible paths give 0.656, 0.379 and 0.932.
- It forbids claiming universality for GR or a derived Hamiltonian for QM.

It also cuts strictly. Frozen operational gates, hashed before execution, eliminate real QM, quaternionic QM, boxworld, polygon theories and all 300 random polygon theories, leaving complex QM. Every hypothesis of the main theorems has a necessity canary:

- Čencov: permutation-only and dimension-scale counterexamples;
- Hessian: Euclidean, total variation and zero counterexamples;
- holonomy: speed, noise, gap and closure counterexamples;
- tameness: the $\sin(1/\theta)$ family.

6 Objection matrix

Objection	Answer	Where
Arbitrariness ("RG picks its gates")	The gates are descent, which Čencov shows forces the geometry. Rigidity gates are hash-frozen before execution, and every hypothesis has a necessity canary	Čencov print; Rigidity
Redundancy ("relabeled statistics")	The four types are pairwise distinct (Proposition 3.1). Untyped practice commits S1–S12	this paper, Section 2
Coincidence ("the data fit by luck")	No claim rests on one fit. The audits are conditional and their assumptions are stated. The structural claims are proofs	Multi-Scale; Structural Closure
Circularity ("assumes QM to get QM")	The premises exclude Hilbert space, the Born rule, tensor products and Schrödinger dynamics; quantum structure appears only as output	Rigidity, C0 freeze
Tautology	The premise is tautological by design, but the theorems are not: Čencov from descent, the Hessian theorem, extension–tomography equivalence, exact thresholds	Sections 1, 4
No novelty ("all classical")	Many components are classical and labelled so. Specific results are claimed (Section 8), and the organizing claim, one typed calculus of ignorance, is offered for refutation	Section 8
Softness	RG forbids specific claims and cuts strictly	Section 5
Unfalsifiable	Explicit falsifiers are given	Section 7
Selection (cherry-picking)	Full denominators, with admitted and abstained designs reported; frozen seeds and hashes; canaries	Spectral; Multi-Scale; batteries

Objection	Answer	Where
Parasitism ("hides behind established systems")	The dependence is deliberate. RG's claims are exactly as strong as the cited theorems, and it makes their residue explicit	throughout

7 How RG would be refuted

- A positive continuous metric family that descends through lossless re-descriptions and is not c·Fisher.
- A monotone, smooth, nondegenerate divergence whose Hessian is not c·Fisher.
- An estimator recovering sign holonomy below the exact threshold $\max(\pi/N, \pi/2 - 2\alpha)$.
- A theory outside the frozen atlas that passes all four rigidity gates and is not complex QM. The imported reconstruction theorems exclude this under their axioms.
- A GR prediction that is not invariant under isometry, or a task-free principle that selects one Petz metric.
- A tame experiment with infinitely many fibre types.

Each of these is a precise mathematical or physical statement, and none is presently known to hold.

8 Novelty ledger

Classical, imported or rederived. Čencov's characterization; Petz's classification; Lovelock; Choquet-Bruhat–Geroch; the hole argument; Cramér–Rao; Bregman/Legendre duality; Gleason/Busch; the reconstructions of QM; Hardt; curve selection; the nerve theorem; Birkhoff–Hopf; Watanabe's RLCT; the Pikovski and Zych decoherence.

Claimed (priority unreviewed).

- Lossless re-descriptions are congruent splittings, so Čencov follows from RG descent.
- The Hessian Čencov theorem and the exact conversion ledger for ignorance.
- Extension–tomography equivalence for transformation quotients.
- The exact noisy holonomy threshold, sharp for every estimator; the minimal analyzer count $n - 1$; non-identifiability of the continuous holonomy.
- Divergence-invariance of wall rates (RLCT) via local comparability; one local forgetting rate; "lossless dynamics never forgets".
- The typed four-kind calculus of ignorance and the GR/QM "class forced, parameters declared" theorem shape.

Organizing claim. No single prior framework, to the author's knowledge, carries all four types of ignorance with their separations, the forced local geometry, and the explicit insufficiency results together. Naming one would refute this claim, and is invited.

9 The corner

A reader who accepts the cited theorems, and cannot refute the twelve exact witnesses, has accepted the following:

- local geometry is forced and insufficient;
- spacetime points are representatives;
- the gravitational and quantum laws are forced as classes and declared as parameters;
- quantum geometry is task-declared;
- global claims from finite data need declared coverage;
- tameness must be declared.

That is the content of RG, under whatever name. The remaining choice is only whether the residue is kept implicit, where it causes S1–S12, or written down.

10 Battery and limits

- `rg_silent_failures.py` runs S1–S12. It is exact or symbolic, with closed forms for S7, S9 and S10, and no random sampling. Reruns are byte-identical.
- Every other number is taken from the companion papers' frozen batteries.
- *Limits.* The dilemmas are only as strong as the cited theorems and the stated hypotheses (tameness, regularity, finite dimension). Priority has not been reviewed. The organizing claim is a claim, offered for refutation.

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